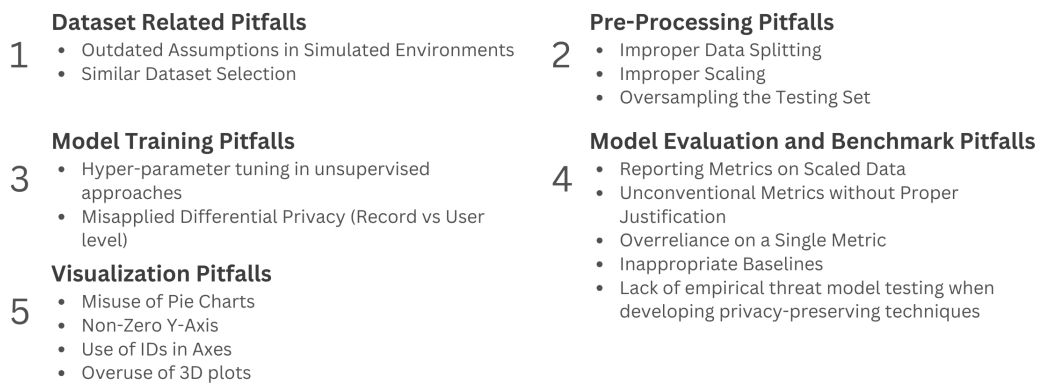

Figure 1: Taxonomy of Machine Learning Pitfalls in Smart Grid Analytics: Structured Identification of Challenges Across Data Preparation, Model Development, and Privacy Preservation. Highlighting critical issues in dataset formulation, preprocessing errors, model evaluation biases, and privacy risks to enhance reliability in energy system applications.

Highlights

Smart Grids and Machine Learning: Uncovering Pitfalls and Improving the Robustness of Models

Emran Altamimi, Hussein Aly, Abdulaziz Al-Ali, Abdulla K. Al-Ali, Qutaibah M. Malluhi

- **Uncovering Critical Pitfalls**: Identify key errors in popular smart grid machine learning research, such as wrong preprocessing and misleading evaluation.
- **Data Preprocessing Flaws**: Demonstrate that improper scaling, splitting, and sampling can lead to information leakage and inflated performance metrics.
- **Misuse in Unsupervised Learning**: Expose the error of tuning hyperparameters with labeled data, which undermines the integrity of the fundamental assumption of unsupervised methods.
- **Streamlined Workflow Recommendations**: Propose best practices to mitigate these pitfalls, ensuring proper real-world applicability.

Smart Grids and Machine Learning: Uncovering Pitfalls and Improving the Robustness of Models

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Abstract

The integration of machine learning into smart grids holds immense promise, yet its practical effectiveness is frequently compromised by overlooked methodological pitfalls. This paper addresses this critical gap by providing a structured analysis of common errors in the smart grid ML development cycle. We identify and demonstrate pitfalls across key stages, including dataset formulation, data preprocessing, and model evaluation. Using illustrative examples from recent literature and reproducible experiments, we show how seemingly minor oversights, such as improper data scaling or the misuse of labels in unsupervised learning, can lead to significantly inflated performance metrics and invalid conclusions. To foster more robust research, this paper presents a framework of best practices and actionable guidelines for practitioners. Our work aims to enhance the validity and reliability of smart grid analytics, ensuring that machine learning models are reproducible, generalizable, and prepared for real-world deployment.

Keywords: Smart grids, Machine learning, Privacy, Data preprocessing, Unsupervised learning

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1. Introduction

Smart grids represent a transformative development in modern energy systems, offering enhanced grid reliability, efficiency, and sustainability. These advanced electrical grids leverage data analytics to optimize the performance of energy distribution and consumption, playing a crucial role in managing and enhancing smart grid operations. They also enable the analysis of vast amounts of data collected from various grid sensors to provide valuable insights, such as load forecasting, anomaly detection, and monitoring the health of the system.

The application of data analytics in smart grids is extensive and diverse (Altamimi et al., 2024) . For instance, demand response programs (O'Connell et al., 2014) rely on analytics to adjust the electricity usage based on supply availability and demand, enhancing grid stability and efficiency. Similarly, the integration of renewable energy sources and the implementation of predictive maintenance strategies are facilitated by sophisticated data-driven techniques that ensure smooth operation and cost efficiency (Weitemeyer et al., 2015).

As the integration of machine learning techniques into smart grids continues to grow, these systems are becoming increasingly capable of making autonomous decisions to improve operational efficiencies. However, this integration introduces new complexities and challenges, particularly in the

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handling and analysis of the data involved. Machine learning models, while powerful, are susceptible to a range of pitfalls that can severely affect their performance and reliability. Common issues include problems with unrealistic dataset assumptions, data preprocessing, inadequate model training, and the use of inappropriate evaluation metrics.

While the potential of machine learning in smart grids is clear, the rush to implementation has often outpaced methodological rigor. This has created a critical research problem: the prevalence of subtle yet significant errors in the machine learning workflow that can invalidate results and mislead the research community. This paper confronts this issue directly by providing a structured analysis of these procedural pitfalls, from flawed dataset assumptions to the misapplication of privacy-preserving techniques. Our central hypothesis is that these common errors lead to overly optimistic performance evaluations and models that are not robust enough for real-world grid operations. By dissecting these issues and offering concrete best practices, we aim to establish a foundation for more reliable, reproducible, and impactful research in the field.

The remainder of this article is structured as follows: Section 2 contextualizes our contribution within existing literature. Section 3 details specific pitfalls across the machine learning workflow, ranging from dataset formulation and preprocessing to evaluation and privacy, supported by experimental demonstrations. Finally, Section 4 presents a framework of best practices, followed by concluding remarks in Section 5.

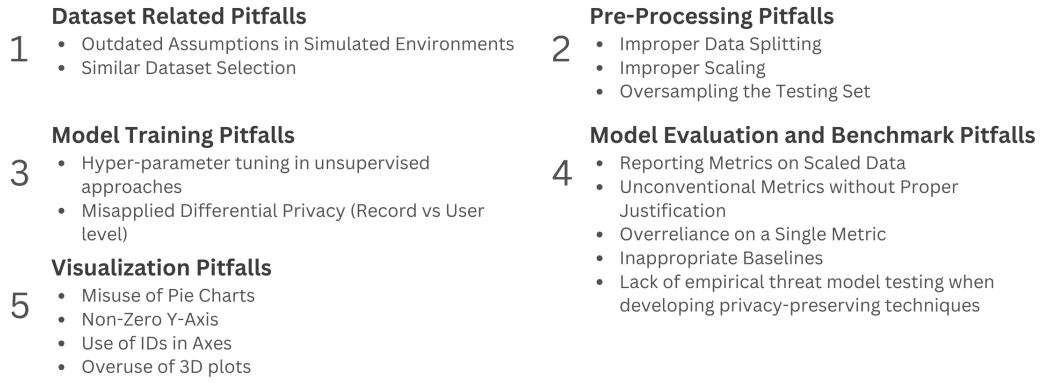

Figure 2: Categories of pitfalls.

2. Related Work

The integration of advanced technologies, particularly machine learning, into smart grids has garnered significant research attention, focusing broadly on technical capabilities, operational efficiencies, and data management challenges. While numerous studies address specific technical hurdles or application domains within smart grids, a systematic examination of the methodological pitfalls inherent in the machine learning development workflow itself remains less explored. Such workflow errors, however, can critically undermine the validity, reliability, and generalizability of machine learning models deployed in these complex energy systems. This section categorizes relevant literature and highlights the gap addressed by our article.

Several studies have investigated practical challenges encountered during smart grid and microgrid implementation. For instance, (Späthe and Conrad, 2022) identified fallacies related to energy measurement assumptions and data processing in microgrid deployments, underscoring the gap between theoretical models and real-world operation. Similarly, (Antonov et al., 2021) provided a valuable analysis of privacy and cybersecurity concerns surrounding Advanced Metering Infrastructure (AMI) systems, primarily from a regulatory and security standpoint. While insightful, these works concentrate on deployment context or security/regulatory frameworks rather than the specific workflow errors made during the machine learning model development process itself.

Research applying machine learning techniques to energy systems often acknowledges data-related difficulties. Recent reviews have begun to touch upon these issues. The work in (Alkhayat et al., 2021), reviewing deep learning for renewable energy forecasting, noted challenges like data scarcity and sampling concerns. Also the work in (Liu et al., 2022) discussed the scalability of machine learning for integrating renewables, emphasizing data heterogeneity and model generalization as key obstacles. Furthermore, (Li et al., 2023) surveyed emerging technologies in smart energy management, identifying potential pitfalls such as model overfitting and the use of non representative datasets. These studies effectively highlight high-level data challenges and machine learning limitations but do not offer a review of the workflow pitfalls that occur throughout the machine learning pipeline, from data preparation to evaluation.

Specific aspects such as machine learning interpretability and privacy, have also received attention. (Machlev et al., 2022) explored the role of

explainable AI (XAI) in power systems, cautioning against misleading visualizations and advocating for better methods to communicate feature importance and uncertainty. Their focus, however, remains centered on model interpretability rather than the broader spectrum of potential errors in the development cycle. In the domain of privacy, (Grataloup et al., 2024) examined the application of federated learning (FL) in energy systems, identifying challenges specific to FL, such as data silos and inconsistent preprocessing across participants. This work underscores the need for robust privacy techniques but concentrates on FL-specific issues rather than general pitfalls in applying privacy-preserving machine learning or other machine learning methodologies.

In contrast to these existing lines of research our work provides a structured analysis focused specifically on the methodological pitfalls within the machine learning development cycle for smart grid applications. We dissect common procedural errors across distinct stages, including dataset selection and problem formulation (e.g., outdated assumptions, inadequate validation diversity), data preprocessing (e.g., information leakage via improper scaling or sampling), model training (e.g., misuse of labels in unsupervised learning), model evaluation and benchmarking (e.g., inappropriate metrics, unfair baseline comparisons), visualization (e.g., misleading chart choices), and the implementation of privacy-preserving techniques (e.g., incorrect application of differential privacy). By identifying and illustrating these workflow-specific errors, often overlooked in broader surveys, this paper aims to provide actionable guidance for improving the rigor, reproducibility, and real-world applicability of machine learning in smart grids. Table 1 provides a comparative summary of the scope of related studies and the specific focus of our contribution.

Table 1: Summary of Related Work and Our Contribution

Article	**Scope**	**Contribution / Focus**
Spathe et al. (2022) (Späthe and Conrad, 2022)	Microgrid deployment challenges.	Discusses data processing and assumption issues in microgrid implementation, highlighting discrepancies between theoretical analysis and real-world deployments.
Antonov et al. (2021) (Antonov et al., 2021)	Privacy and cybersecurity in AMI systems.	Offers legal and regulatory insights regarding AMI security and privacy, without focusing on ML methodology flaws.
Alkhayat et al. (2021) (Alkhayat et al., 2021)	Deep learning for renewable forecasting.	Identifies specific data challenges (e.g., scarcity, sampling) in forecasting, but does not examine broader ML workflow errors.
Machlev et al. (2022) (Machlev et al., 2022)	Explainable AI (XAI) in power systems.	Addresses interpretability and visualization issues for XAI methods, not the systematic pitfalls across the entire ML cycle.
Liu et al. (2022) (Liu et al., 2022)	Machine learning for renewable integration.	Highlights high-level challenges like data quality, heterogeneity, and model generalization in the context of renewables.
Li et al. (2023) (Li et al., 2023)	Emerging technologies in smart energy management.	Points out specific modeling issues like overfitting and dataset bias within energy management applications.
Grataloup et al. (2024) (Grataloup et al., 2024)	Federated learning in energy systems.	Discusses pitfalls specific to FL implementation, such as data silos and preprocessing inconsistencies, not general ML workflow errors.
Our Work	Methodological pitfalls in smart grid ML applications.	Identifies and illustrates recurring procedural errors in smart-grid ML across: dataset selection, preprocessing, training, evaluation, visualization, and privacy implementation.

3. Pitfalls

To investigate common pitfalls, we drew on archetypal case studies across the smart-grid application areas identified in our prior survey (Altamimi et al., 2024): load forecasting, energy-theft and anomaly detection, consumption clustering, electric-vehicle charging analytics, and privacy-preserving learning. Our queries on IEEE Xplore and Web of Science combined each application area with machine-learning method terms (e.g., regression, classification, clustering, anomaly detection, federated learning, and differential privacy), limited to the past five years. We then retained articles appearing in Q1 or Q2 journals by SJR quartile and with at least 30 Google Scholar citations as of September 2025. As illustrated in Fig. 3, this process yielded 275 candidate records.

From these candidates, we selected 25 studies that clearly illustrate one or more of the pitfalls examined here, and retained them for detailed methodological review and qualitative synthesis. We treat these as archetypal case studies because they represent recurring methodological pitfalls across multiple studies in the same application area rather than best- or worst-in-class examples.

Throughout this paper, we cite these 25 articles not to critique specific authors, but to illustrate methodological issues that can systematically undermine the reliability of smart-grid operations.

We then qualitatively assessed these studies, mapping observed procedural flaws onto a conventional machine learning workflow. The identified pitfalls are categorized into the key workflow stages outlined in Fig 2. In the following subsections, we define each issue, illustrate it with examples from recent literature, and, where applicable, demonstrate its impact through experimental results.

3.1. Dataset-Related Problems

The development of machine learning models in smart grid analytics is often hindered by several dataset-related issues, including outdated simulation environments, impractical problem formulation, and narrow dataset selection for validation. These challenges compromise the reliability of insights derived from such data, posing significant risks when these insights inform infrastructural or operational decisions.

One notable example is in studies focusing on electric vehicles (EVs), where researchers often rely on outdated charging parameters. For instance, some popular EV studies from the last three years still utilize 3.3 kW for

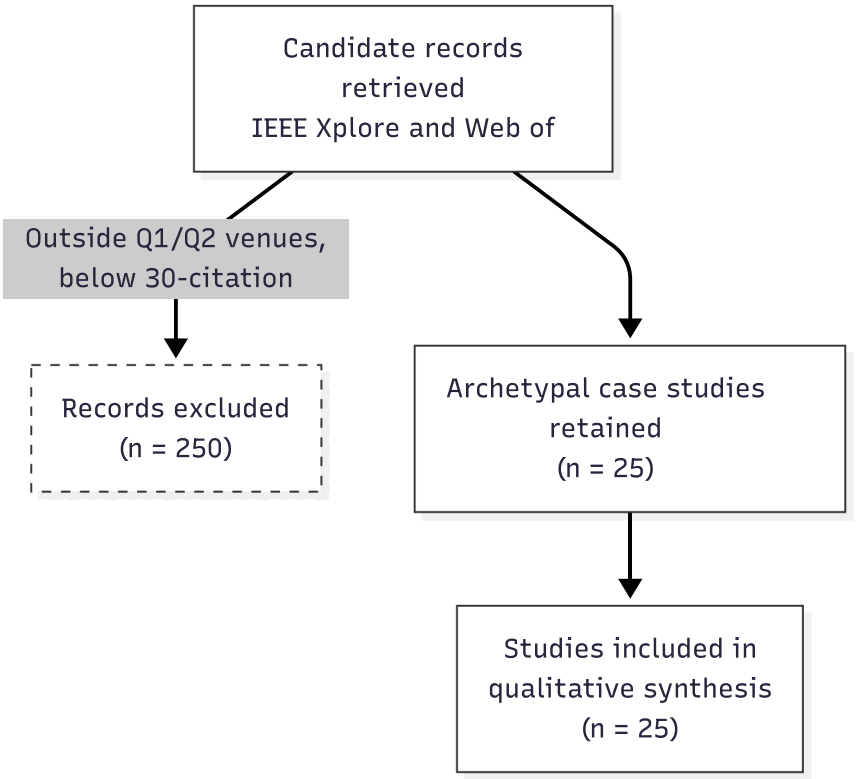

Figure 3: Study selection process. From an initial pool of 275 records identified through queries on IEEE Xplore and Web of Science, venue and citation filters followed by selection for methodological fit retained 25 studies for detailed methodological review.

AC at-home charging and 100 kW for DC fast charging stations (Hu et al., 2022; Lu et al., 2022; Ye et al., 2022; Liu et al., 2022; Liang et al., 2023). By contrast, today's EV charging landscape has evolved considerably. AC at-home charging speeds typically range between 7 kW and 22 kW, with 11 kW being a common standard. Meanwhile, DC fast chargers range from 100 kW up to 350 kW: for example, Tesla's Supercharger V3 can deliver up to 250 kW, and networks such as Electrify America (Electrify America, 2024) and Ionity (IONITY, 2024) offer chargers capable of 350 kW. Premium EV models like the Porsche Taycan and Audi e-tron GT are already equipped to utilize these ultra-fast chargers.

The rapid growth of high-power chargers and compatible vehicles has set new benchmarks for charging speeds. As of 2024, charging rates below 100 kW for public stations are considered outdated except in niche scenarios (U.S. Department of Energy, Alternative Fuels Data Center, 2024; International Energy Agency, 2024), and household chargers under 11 kW are also increasingly rare. Relying on lower-speed charging assumptions in datasets can therefore skew model evaluations and undermine real-world applicability.

Another critical issue involves the choice and characteristics of datasets used to validate proposed models. While it is essential to test models on different datasets to assess generalization, selecting datasets with nearly identical properties can produce misleading results. For example, synthetic energy theft perturbations (often called synthetic attack models) are frequently tested on the Ausgrid (aus, 2020) and Customer Behavior Trials (CBT) (for Energy Regulation , CER) datasets, both of which have a 30-minute sampling rate (Altamimi et al., 2024). Since synthetic attack models typically operate based on the frequency or number of readings (Chuwa and Wang, 2021), using datasets with identical sampling rates does not adequately challenge the model. The model may simply learn patterns tied to that particular rate, appearing to generalize when in reality it does not.

To illustrate, we conducted experiments using an XGBoost classifier and synthetic attack models to detect daily energy theft on three datasets: Ausgrid (aus, 2020) and CBT (for Energy Regulation , CER) (both 48 samples/day, i.e., every 30 minutes), and the UCI electricdiagrams20112014 electric power consumption dataset (Trindade, 2015) (96 samples/day, i.e., every 15 minutes). The classifier produced similar F1 scores (about 0.85) on Ausgrid and CBT, but performed significantly better on the UCI dataset with an F1 score of 0.92. Downsampling the UCI data to 30-minute intervals dropped performance to 0.86, putting it on par with the other two datasets.

These findings suggest that the model leverages sampling-rate-specific patterns rather than truly generalizable features. Consequently, evaluating models solely on datasets with similar sampling rates can overestimate their real-world performance.

We therefore recommend testing models on datasets with varying sampling frequencies and other differing attributes. Detailed experimental setups, including clear explanations of dataset characteristics, can further enhance reproducibility and robustness. By carefully aligning simulation assumptions with real-world conditions and validating models on appropriately diverse datasets, researchers can ensure that their findings are both reliable and broadly applicable.

3.2. Pre-processing Pitfalls

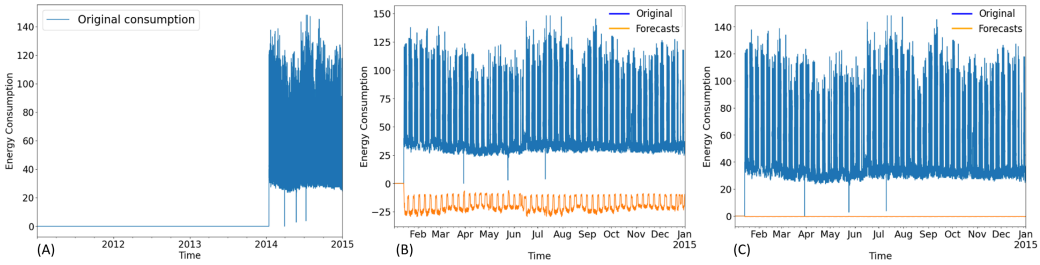

Figure 4: (A) Original consumption of a sampled anomalous user, (B) Wrong scaling forecasts, (C) Correct scaling forecasts

The sequence of data preprocessing steps is vital for preserving the integrity of model validation in smart grid analytics. One fundamental principle is to divide the data into training, validation and testing sets prior to applying any preprocessing techniques. This step is critical to prevent information leakage from the testing set into the training process. However, this principle can sometimes be inadvertently violated, resulting in unreliable model performance results. This section addresses several prevalent pre-processing pitfalls in smart grid analytics, including improper data splitting and scaling, and the incorrect application of sampling techniques like oversampling and undersampling on the testing set.

A subtle error arises in the context of forecasting energy consumption, where the data split is often done along the temporal dimension rather than by individual examples or users. Given that the dataset typically consists of a single feature (energy consumption), researchers sometimes mistakenly apply

scaling techniques across the entire dataset, including the testing set. This strategy, employed in recent works such as (Guo et al., 2022) and (Liang and Tang, 2022), introduces data leakage, thereby inflating model performance metrics (Ahsan et al., 2021) or resulting in unrealistic and nonsensical results in other cases, especially in personalised forecasting models. To illustrate this, we conducted an experiment using an LSTM-based forecasting model to a sample real-world anomalous consumer from the UCI dataset (Trindade, 2015) (see Figure 4 a). Following common convention, the testing data is set to be the last year (i.e., the year 2014). In the first scenario, we incorrectly applied Min-Max scaling to the entire consumption profile before splitting, which caused the model's predictions to be consistently below zero in energy usage (see Figure 4 b). In the second scenario, we correctly applied scaling only to the training set and then transformed the test set using the same scaler fitted on the training data. This approach preserved the integrity of the model's predictions, aligning closely with the real consumption patterns and accurately reflecting the variability in the data (see Figure 4 c). These results demonstrate that improper scaling can severely impact model performance, leading to unrealistic and nonsensical predictions, and highlight the importance of applying preprocessing steps after dividing the data to prevent information leakage.

Another preprocessing pitfall involves the improper application of examples sampling techniques like oversampling and undersampling. While preprocessing the testing set is necessary, it is incorrect to apply these sampling methods to it, as these sampling techniques aim to balance the data, often favoring the minority class, which skews the results and artificially inflates performance metrics by often testing on copies of the testing data. For example, when applying an XGBoost model (Punmiya and Choe, 2019) to detect theft on the SGCC dataset (Zheng et al., 2018), the impact of oversampling the testing set can be dramatic. Oversampling the testing set drastically skews the results and provides a deeply unrealistic and flawed perception of the model's performance, as illustrated in Figure 5.

Consequently, the modified testing set no longer represents the true data distribution, leading to misleadingly high performance metrics (Mohammed et al., 2020). This issue is highlighted in studies like (Qu et al., 2020) and (Takiddin et al., 2022), where dramatic performance improvements were observed due to unrealistic assumptions in this approach.

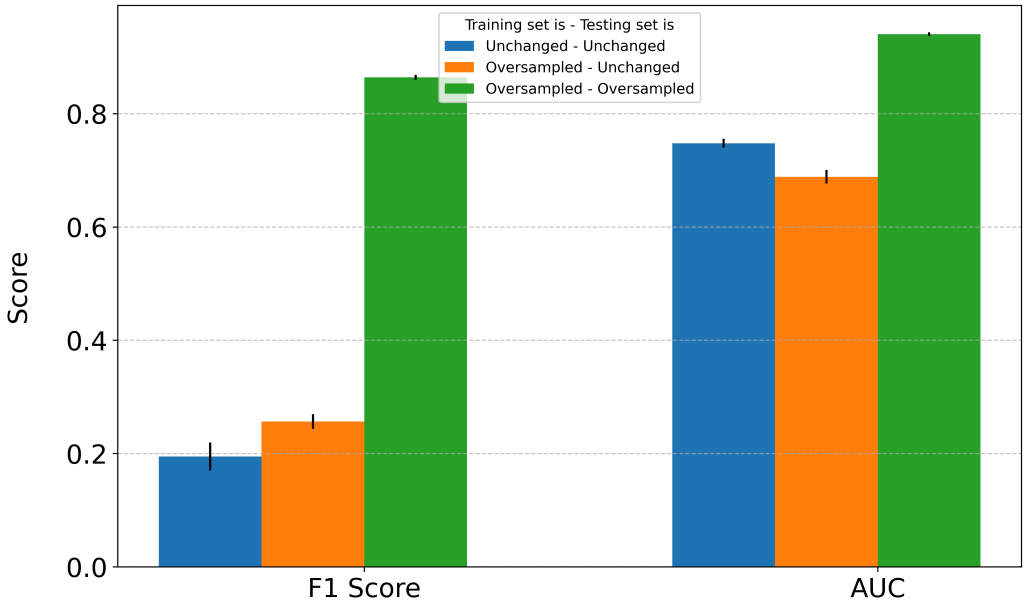

Figure 5: Effect of oversampling on the testing set for XGBoost model performance on SGCC dataset.

3.3. Model Training Pitfalls

While model training is generally well-executed in the literature, a subtle issue arises with unsupervised approaches. Specifically, using actual labels to tune hyper-parameters, such as the number of clusters in clustering algorithms or the optimal threshold, contradicts the unsupervised nature of these methods. For instance, although the studies in (Wang et al., 2019) and (Viegas et al., 2018) appropriately fine-tune parameters on the training set and report results on the testing set, their methodology undermines the unsupervised premise. If utility providers do not have labels of previous theft incidents, then tuning the hyper-parameters in this fashion becomes impossible. The difference in performance due to tuning can be staggering. For example, the performance of Wang's approach (Wang et al., 2019) to detect synthetic theft attacks, with and without tuning the two main parameters of their approach: the anomaly score for each reading and the number of anomalous readings in a day required to label the day as theft. The approach is evaluated on two different datasets using the forecasting residuals of two different forecasting models. The huge variations in performance is

illustrated in Table 2.

Table 2: The accuracy of Wang's (Wang et al., 2019) approach with and without tuning the parameters. The forecasting residuals used are of two forecasting models and two datasets.

Dataset	UCI Electricdiagram		Ausgrid	
Model architecture	LSTM	ARIMA	LSTM	ARIMA
With parameter tuning (unsupervised assumption violated)	0.79	0.80	0.65	0.71
Without parameter tuning (unsupervised assumption maintained)	0.502	0.509	0.5	0.503

To keep the unsupervised premise intact, we recommend using intrinsic, label-free validation criteria and sensitivity analyses rather than tuning hyperparameters with labeled data. The choice of criterion should depend on the structure of the unsupervised task. For clustering, internal validity indices such as Silhouette or Davies-Bouldin, together with stability under perturbation, can be used to assess whether partitions remain coherent without external labels (Arbelaitz et al., 2013; Meinshausen and Bühlmann, 2010). For reconstruction-based anomaly detectors, such as autoencoder approaches, threshold selection can instead be guided by the stability or separation of reconstruction-error distributions rather than by labeled attack examples (Ruff et al., 2021; Takiddin et al., 2022). In threshold-based anomaly detection settings where outputs are naturally represented as flagged sets, such as identifying energy theft from forecasting residuals, we use the stability of the flagged anomaly set across a grid of threshold values as one concrete illustration of this broader principle (Ruff et al., 2021; Meinshausen and Bühlmann, 2010).

Mathematically, let A_θ represent the set of consumers or readings flagged as anomalous given a hyperparameter configuration θ (e.g., a specific residual threshold). If the parameter is perturbed by a small step to $\theta + \Delta\theta$, the stability of the model can be quantified using the Jaccard similarity index between the resulting anomaly sets:

$$J(A_\theta, A_{\theta+\Delta\theta}) = \frac{|A_\theta \cap A_{\theta+\Delta\theta}|}{|A_\theta \cup A_{\theta+\Delta\theta}|}$$

Algorithmically, researchers should iteratively compute J across a realistic parameter grid. A highly sensitive threshold will yield a low Jaccard index, indicating that the model's decisions are erratic and highly dependent on arbitrary tuning. Conversely, identifying a region where $J \approx 1$ reveals a stable operational threshold where the unsupervised model consistently isolates the same anomalous distribution. This quantitative approach is one example of an intrinsic, label-free validation strategy and is particularly suitable when the output of the unsupervised method can be represented as a set of flagged observations. More broadly, the key principle is to match the intrinsic validation criterion to the model family and output structure while avoiding the use of external labels.

Another pitfall arises when the privacy unit is defined too narrowly for smart-grid time-series data. Because each consumer contributes a correlated trajectory of hourly or daily readings, protecting individual timestamps does not by itself guarantee that the inclusion or exclusion of a consumer's full load profile has limited influence on the released model. Studies such as (Dai et al., 2023) and (Khalil et al., 2023) discuss differential privacy in federated smart-grid settings, but the methodological question is whether one consumer's entire local dataset can be removed or replaced while the mechanism's output remains nearly unchanged (Ponomareva et al., 2023). A recent critical review (Blanco-Justicia et al., 2022) makes precisely this point at the methodological level, observing that when client records are highly correlated, as smart-meter readings are, local differential privacy protects individual records (instance-level privacy) but does not hide the client's overall contribution (client-level privacy).

However, protecting individual readings, or discussing privacy only at the level of intermediate local updates, is not sufficient to establish user-level privacy. In time-series smart-grid applications, the protected unit should instead be defined as one consumer's full trajectory over the relevant reporting window, because that full trajectory is what links the data back to the user.

A better practice is therefore to bound each user's contribution before aggregation, then add noise to the aggregated statistic or model update rather than to each reading independently. When the same users participate across multiple training rounds, the cumulative privacy loss should also be reported via the resulting (ϵ, δ) guarantee together with the participa-

tion assumptions, so that the privacy claim matches the deployment setting (Ponomareva et al., 2023; Near et al., 2023).

3.4. Model Evaluation and Benchmarking Pitfalls

Accurately evaluating model performance is crucial, yet certain pitfalls can significantly distort the perceived efficacy of a model, leading to misleading conclusions. This section discusses several pitfalls in model evaluation and benchmarking in smart grid analytics, including reporting forecasting results on scaled data, unconventional reporting methods, over-reliance on a single evaluation metric, neglecting performance on minority classes, and the use of inappropriate baselines.

Scaling data samples is a crucial step in the model construction pipeline of smart grid applications. However, when not handled carefully, it can lead to serious misinterpretation of results. For example, in works like (Fernández et al., 2022), results were reported before appropriately scaling back the forecasted data. This premature reporting can create a false impression of the model's accuracy, making it difficult for other researchers to fairly compare these results with those from other studies, thereby undermining the validity of the conclusions drawn.

The use of unconventional methods to report results can further complicate the assessment of a model's performance. For instance, the work in (Fenza et al., 2019) presented forecasting results using only the maximum and minimum Root Mean Squared Error (RMSE) values. While such an approach might highlight extremes that are useful for specific applications like anomaly detection, it fails to provide a comprehensive picture of the model's overall performance. This selective reporting can obscure critical weaknesses or to assess the model's consistency across different scenarios.

Although RMSE is a widely used metric, relying on it alone in datasets with significant variance, such as those that include both household and industrial energy data, can be misleading. High variance can skew RMSE, masking how well the model performs across different types of data. For instance, reporting Mean Absolute Percentage Error (MAPE) in the work of (Ertugrul, 2016) would have provided a more balanced evaluation and insights into performance across different scales. However, MAPE is also problematic in datasets with values close to zero, such as periods of zero energy consumption, as it can result in disproportionately high errors due to division by zero or near-zero values, which skews the evaluation. To minimize the risks associated with using specific measures, future work should use a

more diverse set of performance assessments.

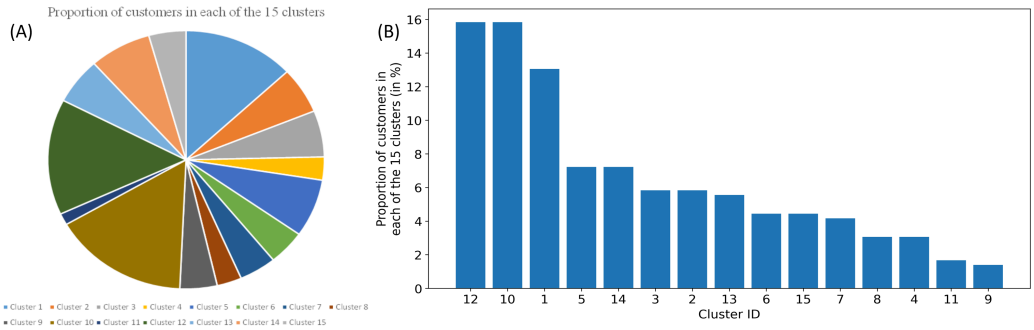

Figure 6: The proportion of consumers in each cluster is visualized in A) a pie chart (Aurangzeb et al., 2021) B) a sorted bar plot

Accurate evaluation of machine learning models also hinges on the careful selection of metrics, efficient computation, and valid benchmarking. For instance, while (Wang et al., 2022) employed a range of evaluation measures, relying primarily on the F1 score for the majority class can obscure shortcomings in minority classes. In the example of detecting EV users, the 3.3 kW chargers achieved an accuracy of 0.95 but an F1 score of only 0.67, while the 6.6 kW chargers recorded an accuracy of 0.97 with an F1 score of 0.8. Such disparities reveal that focusing on a single class or metric provides an incomplete view of overall performance, especially in imbalanced datasets.

Beyond predictive accuracy, it is crucial to assess the efficiency of machine learning solutions. Evaluating efficiency in terms of computational resource demands, processing time, and scalability ensures that a model is not only accurate but also practical for real-world deployment. A highly accurate model may still be impractical if it requires excessive resources or incurs prohibitive latency, particularly in settings where computational capacity is limited. Incorporating efficiency metrics bridges the gap between theoretical performance and operational feasibility.

A further challenge arises from the use of inappropriate baselines in benchmarking experiments. For example, (Hasan et al., 2019) referenced the results of previous studies without replicating those models in the same data set, which can lead to misleading comparisons due to the inherent differences of the data set. Ensuring that baseline models are evaluated under the same conditions is essential to establish a fair and meaningful performance comparison.

Another critical issue arises when the effectiveness of privacy techniques is not empirically validated under an explicit threat model. Studies such as (Gough et al., 2021) and (Liu et al., 2020) implemented DP without thoroughly testing the approach against potential privacy attacks, with the justification that DP is mathematically guaranteed. However, the choice of ϵ, the privacy budget, significantly influences the level of privacy preserved, and without proper evaluation, it is difficult to assess whether the applied mechanism genuinely protects consumer privacy and, more crucially, to what level. Failing to do so can result in overconfidence in the privacy guarantees provided, potentially leaving the system vulnerable to privacy breaches. As (Blanco-Justicia et al., 2022) show empirically, implementations that rely on DP relaxations paired with loose privacy parameters (for example, ϵ values well above the $\epsilon \leq 1$ range recommended by DP's originators) frequently deliver practical protection no stronger than plain noise addition, leaving the actual level of protection to be established *ex post* against concrete attacks. Empirical attacks such as membership inference are not substitutes for formal DP guarantees. Their role is to complement formal privacy analysis by stress-testing the implemented mechanism under plausible adversarial settings.

3.5. Visualization Pitfalls

Effective visualization is crucial for conveying insights clearly and accurately, yet common pitfalls can distort the message or mislead the audience (Bresciani and Eppler, 2015). This section addresses three prevalent pitfalls observed in smart grid analytics and other related fields: the use of pie charts, figures with non-zero Y-axis limits, and the use of serial numbers or IDs instead of distribution plots.

Pie charts are often criticized for their inability to accurately represent data, especially when dealing with more than a few categories. The primary issue is that humans struggle to compare angles and areas, making it difficult to perceive differences in the size of the pie slices accurately. This can lead to misinterpretation of the data, particularly when slices are similar in size. In smart grid analytics, where precise understanding of data is crucial, relying on pie charts can result in poor decision-making. For example, studies such as (Ryu et al., 2020a) and (Aurangzeb et al., 2021) used pie charts to represent data distributions across multiple categories, leading to potential misinterpretations due to the challenges in comparing the relative sizes of the slices, as in Figure 6 reported by (Aurangzeb et al., 2021). In-

stead, bar charts or stacked bar charts are recommended, as they allow for more straightforward comparisons and can accommodate a larger number of categories more effectively. For example the pie chart in (Aurangzeb et al., 2021) shown in Figure 6 (a) could be better represented in a sorted bar plot as shown in Figure 6 (b).

When plotting measured performance scores and errors, starting the Y-axis at a value other than zero can exaggerate the perceived differences between data points, leading to potentially misleading interpretations. This practice, while sometimes used to highlight small differences, can distort the true scale of variation, especially in performance metrics with a known range such as precision, recall, F1 score, accuracy, and so on, which are central to model evaluation in smart grid analytics. For instance, the works by (Ryu et al., 2020b), (Cui et al., 2022), and (Wang et al., 2020) presented figures with non-zero Y-axis baselines that caused the differences between data points to appear more pronounced than they are, potentially leading to an overestimation of the model's performance. To maintain the integrity of the visual representation, it is generally recommended to start the Y-axis at zero. If a non-zero baseline is necessary, it's important to include a figure that starts from zero alongside a zoomed-in subplot. This approach provides full context while allowing detailed comparisons, helping to prevent misinterpretation.

Another pitfall is the use of serial numbers or IDs in place of distribution plots. While these identifiers might be relevant for specific cases, they often do not convey the underlying data distribution, which is critical for understanding trends, outliers, and variations within the dataset. For instance (Aurangzeb et al., 2021) used IDs to report outliers without showing the distribution of the data, however important insights about the overall consumption behavior would be easier to notice with other visualization techniques. For example, while the authors in (Aurangzeb et al., 2021) utilized IDs to visualize the outliers in each consumer as shown in Figure 7 (A), the use of a histogram for the same data makes it easier to note insights about the data as shown in Figure 7 (B).

The use of 3D plots generally conveys more information in figures and is useful in numerous cases. However, the added dimension distorts the perception of the data, making it difficult to accurately interpret values, compare data points, or discern relationships between variables. This issue is particularly problematic in cases where precise data interpretation is necessary. Studies such as (Feng et al., 2020), (Lin et al., 2019), and (Liang and Ma,

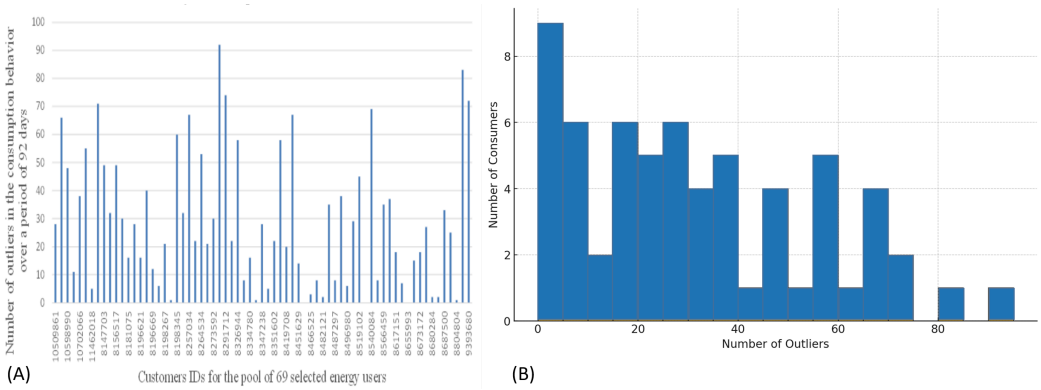

Figure 7: The number of outliers in 92 days of readings for 69 consumers is visualized using (A) Customer IDs (Aurangzeb et al., 2021) (B) a histogram.

2021) utilized 3D plots, leading to potential misinterpretations due to the challenges in accurately reading and comparing data across the added dimension. For example, the 3D plot presented by (Feng et al., 2020) shown in Figure 8 (A) depicts how Elasticity affects Time of Use (ToU) price rates across a single day takes a little more effort from the reader compared to a simple 2D plot with several plot lines as shown in Figure 8 (B). As an alternative, 2D plots, such as heat-maps, scatter plots, or contour plots, are often more effective, providing clearer insights without the confusion introduced by the third dimension.

4. Discussion and Best Practices

To systematically address the issues identified in this study, it is essential to distinguish between pitfalls that are inherent to general machine learning methodology and those specific to the smart grid domain. As summarized in Table 3, General ML Pitfalls, such as data leakage or misleading visualization, are domain-agnostic but result in dangerously inflated performance metrics that can mislead grid operators. In contrast, Domain-Specific Pitfalls stem from a disconnect between the model and the physical reality of the energy grid, such as outdated EV charging assumptions, the misapplication of privacy units to time-series energy data or unrealistic energy theft synthetic data generation. Addressing both categories is required to ensure that models are not only statistically sound but also physically relevant.

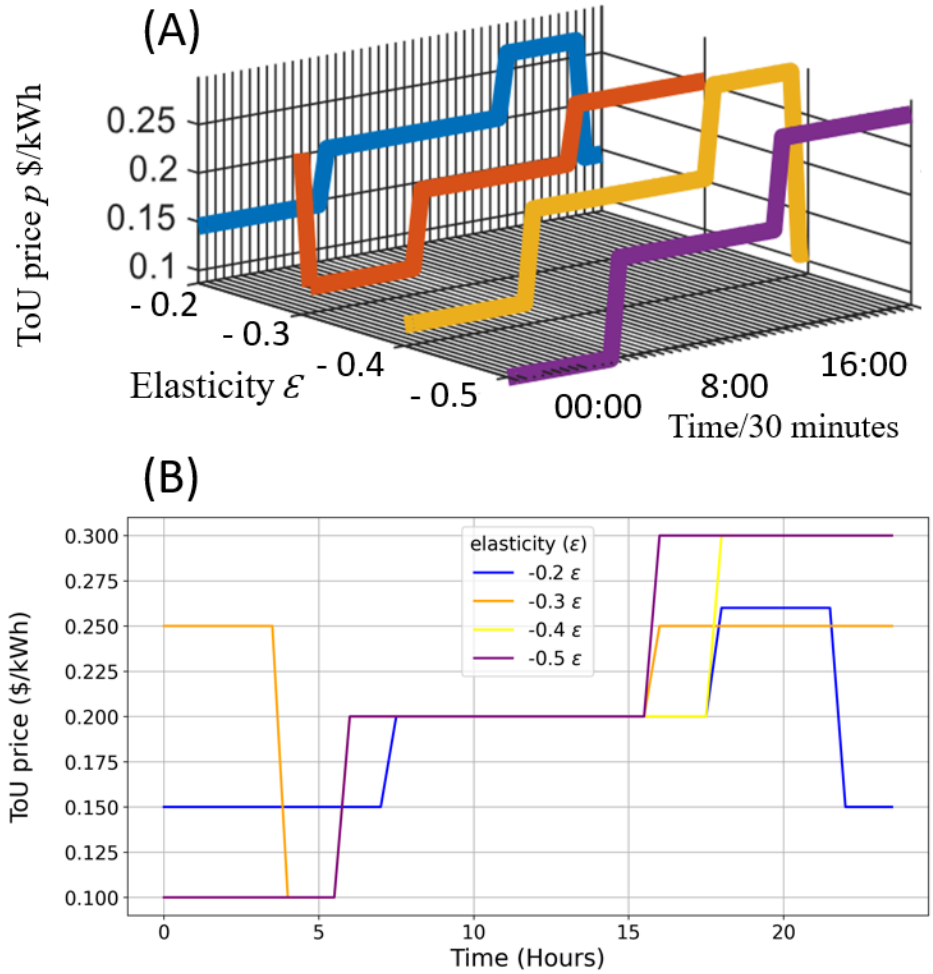

Figure 8: The effect of Elasticity on Time of Use (ToU) tariffs visualized in (A) 3D plot (Feng et al., 2020), and (B) 2D plot.

Table 3: Classification of Pitfalls: Methodological Integrity vs. Domain Specific

Workflow Stage	**Nature of Pitfall**	**Description of Issue**
Dataset Formulation	**Domain Specific**	Reliance on outdated grid parameters (e.g., old EV charging rates) or physically unrealistic attack scenarios.
Preprocessing	**Methodological Integrity**	Data leakage (e.g., scaling before splitting) and improper sampling of time-series testing data.
Model Training	**Mixed**	*Methodological Integrity:* Tuning unsupervised models with labels. *Domain:* Misapplying Differential Privacy to individual records rather than user profiles.
Evaluation	**Methodological Integrity**	Reliance on single metrics (e.g., accuracy on imbalanced data) or unfair baselines.
Visualization	**Methodological Integrity**	Misleading charts (e.g., non-zero axes, 3D plots) that obscure data distribution.

In light of the pitfalls outlined in the previous sections, this section proposes a recommended workflow for applying machine learning techniques in the context of smart grids. The workflow is summarized in Table 4, which highlights common pitfalls at each stage and provides corresponding best practices to mitigate them. By adhering to these recommendations, researchers and practitioners can improve the reliability and generalizability of their models, ensuring that the insights derived from smart grid data are both accurate and actionable.

Table 4: Common Pitfalls and Recommended Best Practices in Machine Learning for Smart Grids

Common Pitfalls	Recommended Best Practices
Dataset Selection & Problem Formulation	
• Outdated or unrealistic assumptions (e.g., old EV charging speeds) • Single dataset reliance • Synthetic attack models not aligned with real data	• Validate simulation parameters against recent, real-world usage • Evaluate models across heterogeneous datasets with different sampling frequencies, noise levels, consumer profiles (and where relevant class imbalance), and report results separately for each dataset to assess robustness • Align synthetic scenario generation with realistic problem definitions
Preprocessing	
• Data leakage from the test set (e.g., fitting scalers on entire dataset) • Oversampling or undersampling on the test set • Failure to account for temporal splits	• Report the split unit, train/validation/test boundaries, and preprocessing order, ensuring splitting occurs **before** any scaling to prevent leakage • Fit scalers only on the training set, then transform validation/test set • Avoid any form of oversampling/undersampling on the validation/test set • Maintain temporal order in time-series data
Model Training	

Continued on next page

Table 4 – continued from previous page

Common Pitfalls	Recommended Best Practices
• Using labeled data to tune hyperparameters in unsupervised tasks • Overfitting through excessive hyperparameter tuning • Misapplied DP unit (record vs. user-level)	• Preserve the unsupervised setting by using intrinsic validation criteria and sensitivity analyses matched to the task • Employ cross-validation and early stopping to reduce overfitting whenever possible • Clearly distinguish between supervised vs. unsupervised objectives • Align DP with the protected unit in time-series data (one consumer's full trajectory, not isolated readings)
Model Evaluation & Benchmarking	
• Reporting results on scaled data only • Using a single or unsuitable metric • Comparing with baselines trained on different datasets • Lack of empirical threat model testing when developing privacy-preserving frameworks	• Invert scaling before final performance reporting where possible • Implement a balanced, multi-metric assessment framework (e.g., RMSE, MAPE, minority class F1-score, and computational efficiency) • Evaluate baselines under the same task definition, split strategy, metric set, and dataset conditions for fair comparison • Highlight performance on minority classes or edge cases • Use empirical attacks (e.g., membership inference) as stress tests that complement formal DP guarantees
Visualization	

Continued on next page

Table 4 – continued from previous page

Common Pitfalls	Recommended Best Practices
• Pie charts for more than a few categories (difficult for comparisons) • Non-zero Y-axis chart limits exaggerating value differences • 3D plots distorting data interpretation	• Replace pie charts with bar or stacked bar charts for clarity • Start Y-axis at zero and provide a zoomed subplot if needed • Use 2D plots (heatmaps, line plots) for better clarity and comparability • Use distribution plots (histograms/violin/box plots) to reveal underlying data structure

The workflow begins by emphasizing realistic problem formulation and dataset selection, guiding the creation of experiments that reflect current grid technologies and usage patterns. It then moves on to address critical preprocessing steps, underscoring the importance of data splitting prior to any transformations to prevent information leakage. Model training considerations are also highlighted, particularly the need to preserve the assumptions of unsupervised methods and to avoid hyper-parameter tuning with labeled data in such settings.

The subsequent phase focuses on model evaluation and benchmarking, recommending the use of multiple metrics to capture different performance aspects and advocating for reproducible baselines on the same datasets. Visualization pitfalls, including misleading axis scales and the overuse of pie or 3D charts, are also mentioned to ensure that reported results are conveyed clearly and accurately. Finally, privacy concerns are addressed through the careful implementation of privacy-preserving techniques and rigorous threat model assessments.

5. Conclusion

This article has brought to light several critical pitfalls in the development of machine learning solutions for smart grid analytics. By examining challenges across the entire development cycle, from dataset selection to model evaluation, we emphasize the importance of vigilance and methodological rigor in this rapidly evolving field.

Recognizing these pitfalls is crucial for researchers and practitioners aiming to enhance the reliability and effectiveness of their models. By being

aware of issues such as unrealistic dataset assumptions, improper data preprocessing, inadequate training practices, and flawed evaluation metrics, professionals can proactively implement strategies to mitigate these challenges. This proactive approach not only improves the validity of individual studies but also contributes to the overall advancement of smart grid technologies.

The insights provided in this review serve as a guide to avoid common mistakes and adopt best practices in machine learning applications within smart grids. By addressing these pitfalls, the community can develop more robust, generalizable, and effective solutions that better meet the complex demands of modern energy systems. Ultimately, fostering such rigorous standards will accelerate innovation and deployment of reliable smart grid technologies, leading to enhanced grid reliability, efficiency, and sustainability.

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